DEPARTMENT OF ARTIFICAL INTELLIGENCE AND MACHINE LEARNING
Choice Based Credit System (CBCS)
SEMESTER – IV

Analysis & Design of Algorithms Lab (0:0:2:0) 1
(Effective from the academic year 2023 -24)

Course Code	**BCDL404**	CIE Marks	50
Teaching Hours/Week (L:T:P:S)	0:0:2:0:0	SEE Marks	50
Total Number of Contact Hours	2 hours	Exam Hours	3 Hours

Course Objectives:
This course aims to prepare the students to:

- To design and implement various algorithms in C/C++ programming using suitable development tools to address different computational challenges.
- To apply diverse design strategies for effective problem-solving.
- To Measure and compare the performance of different algorithms to determine their efficiency and suitability for specific tasks.

SL.NO	Experiments
1.	Sort a given set of n integer elements using Selection Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator. Demonstrate using C/C++, how the brute force method works along with its time complexity analysis: worst case, average case and best case.
2.	Sort a given set of n integer elements using Quick Sort method and compute its time complexity. Run the program for varied values of n> 5000 and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator. Demonstrate using C/C++ how the divide-and-conquer method works along with its time complexity analysis: worst case, average case and best case.
3.	Sort a given set of n integer elements using Merge Sort method and compute its time complexity. Run the program for varied values of n> 5000, and record the time taken to sort. Plot a graph of the time taken versus n. The elements can be read from a file or can be generated using the random number generator. Demonstrate using C/C++how the divide-and-conquer method works along with its time complexity analysis: worst case, average case and best case.
4.	Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Kruskal's algorithm.
5.	Design and implement C/C++ Program to find Minimum Cost Spanning Tree of a given connected undirected graph using Prim's algorithm.
6.	Design and implement C/C++ Program to find shortest paths from a given vertex in a weighted connected graph to other vertices using Dijkstra's algorithm.
7.	Design and implement C/C++ Program to obtain the Topological ordering of vertices in a given digraph.
8.	Design and implement C/C++ Program to solve 0/1 Knapsack problem using Dynamic Programming method.

9.	Design and implement C/C++ Program to find a subset of a given set S = {sl , s2,. .. ,sn} of n positive integers whose sum is equal to a given positive integer d.
10.	Design and implement C/C++ Program for N Queen's problem using Backtracking.

Course outcomes (Course Skill Set):

At the end of the course the student will be able to:

CO1: Develop programs to solve computational problems using suitable algorithm design strategy.

CO2: Compare algorithm design strategies by developing equivalent programs and observing running times for analysis (Empirical).

CO3: Make use of suitable integrated development tools to develop programs.

Assessment Details (both CIE and SEE)

Lab Evaluation Scheme

1. Ten marks for every experiment (10 X 10 = 100 marks), round it off to ***30 marks***.
2. Ten marks for every experiment will be evaluated for write-up, program execution, the procedure followed while execution and viva voce after each exercise.
3. Internal practical test for 100 marks to be given and the marks scored will be scaled down to ***20 marks***.
4. A Minimum of ***20 mark*** is to be scored in CIE.
5. SEE examination for the Lab is to be conducted for 100 marks and reduced to ***50 marks***.
6. A Minimum of ***18 marks*** is to be scored in SEE.

Note: Open Ended experiment will be done by the students in the Lab session. A total mark of 40 is to be scored by the student from both CIE and SEE together out of 100.

Semester End Evaluation (SEE):

- SEE marks for the practical course are 50 Marks.
- SEE shall be conducted jointly by the two examiners of the same institute, examiners are appointed by the Head of the Institute.
- The examination schedule and names of examiners are informed to the university before the conduction of the examination. These practical examinations are to be conducted between the schedule mentioned in the academic calendar of the University.
- All laboratory experiments are to be included for practical examination.
- (Rubrics) Breakup of marks and the instructions printed on the cover page of the answer script to be strictly adhered to by the examiners. **OR** based on the course requirement evaluation rubrics shall be decided jointly by examiners.
- Students can pick one question (experiment) from the questions lot prepared by the examiners jointly.
- Evaluation of test write-up/ conduction procedure and result/viva will be conducted jointly by examiners.
- General rubrics suggested for SEE are mentioned here, writeup-20%, Conduction procedure and result in -60%, Viva-voce 20% of maximum marks. SEE for practical shall be evaluated for 100 marks and scored marks shall be scaled down to 50 marks (however, based on course type, rubrics shall be decided by the examiners)
- Change of experiment is allowed only once and 15% of Marks allotted to the procedure part are to be made zero.

The minimum duration of SEE is 02 hours

Suggested Learning Resources:

Virtual Labs (CSE): http://cse01-iiith.vlabs.ac.in/